



Quantitative Investing: Skill, Risk, and the Logic of Active Management

SMM921: Trading and Microstructure

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$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Objectives and structure of the lectures

- **Lecture 1:** Foundations and performance measurement
- **Lecture 2:** Forecasting and portfolio construction
- **Lecture 3:** Implementation, extensions, and introduction to dynamic allocation

Reading

For today's material read:

- Ang, *Asset Management: A Systematic Approach to Factor Investing*, Chs. 2 and 3
- Grinold and Kahn, *Active Portfolio Management*, Chs. 4 and 5
- Grinold and Kahn, *Advances in Active Portfolio Management*, Chs. 2 and 3
- Kinlaw, Kritzman and Turkington, *A Practitioner's Guide to Asset Allocation*, Chs. 1, 2 and 3

Basic paradigm I

The basic **mean-variance** paradigm for investing:

$$Utility = mean\ return - risk\ penalty - costs$$

Basic paradigm II

The basic **mean-variance** paradigm for investing:

Utility = mean return – risk penalty – costs

$$U = \mathbf{w}'\boldsymbol{\mu} - \frac{\lambda}{2} \mathbf{w}'\boldsymbol{\Sigma}\mathbf{w} - \boldsymbol{\tau}'|\mathbf{w} - \mathbf{w}_0| \quad \text{s.t.} \quad \mathbf{1}'\mathbf{w} = 1$$

where $\mathbf{w}_0$ and $\mathbf{w}$ are the initial and final portfolio weight vectors, $\boldsymbol{\mu}$ is a vector of expected returns, $\boldsymbol{\Sigma}$ is the return covariance matrix and $\boldsymbol{\tau}$ is a vector of trading costs. λ is the coefficient of risk aversion. There are N assets.

Every single input matters

A good portfolio manager does the following:

- **Forecasts returns well** (μ is correlated with realised return outcomes)
- **Understands risk** (Σ is estimated accurately)
- **Understands tolerance for risk** (knows λ)
- **Has costs under control** (has estimated τ and so knows how much trading each asset will cost)

Questions that arise: risk and risk aversion

- Is variance the right measure of risk?
- How can I interpret a risk aversion coefficient?
- How should I estimate risk aversion?
- Is it constant over time?
- Does it change as an investor ages or gets wealthier?

Questions that arise: return forecasts

- What models or rules allow one to make non-trivial return forecasts?
 - e.g. “Value and Momentum Everywhere” [▶ Published paper](#)
- How can I characterise the quality of my forecast?
- How does forecast quality translate into investment performance?
- How should I combine forecasts?
- Can the market tell me something about future returns?

Questions that arise: covariance matrices

- What is the best way to estimate variances and covariances of returns?
 - Simple historical sample estimation can be poor
- Can we use statistical techniques to improve covariance estimation?
 - Factor models
 - Bayesian methods (e.g. shrinkage)

Questions that arise: trading costs

- Which assets are likely to be cheapest to trade, and when?
 - This requires estimation of **liquidity**
- How does liquidity interact with the ability to forecast returns?
- What factors explain asset liquidity?
 - Information risk and price risk
- Which institutional and structural factors have caused liquidity to change in recent years?
 - Algorithmic trading, HFT, dark pools, Dodd-Frank, MiFID II, etc.

Basics: preferences and objectives

We start with a standard **mean-variance preference** set-up. The investment manager solves:

$$\max_{\mathbf{w}} \mathbf{w}'\boldsymbol{\mu} - \frac{\lambda}{2}\mathbf{w}'\boldsymbol{\Sigma}\mathbf{w} \quad \text{s.t.} \quad \mathbf{1}'\mathbf{w} = 1$$

This is:

- Common in practice
- Simple to deal with analytically

MV utility: usage and issues

But is the use of MV utility sensible?

- For MV to be the best approach, we need either:
 1. Returns to be **normally distributed**, or
 2. Investors to have **quadratic utility** (which implies increasing absolute risk aversion)

But:

- Returns are often not normally distributed – tail outcomes occur more frequently than a normal would predict
- Wealthy investors are usually willing to take more risk
- Investors dislike losses more than they enjoy similarly sized gains
- Investors care about skewness (and potentially higher moments) too

MV utility: a special case of a more general framework

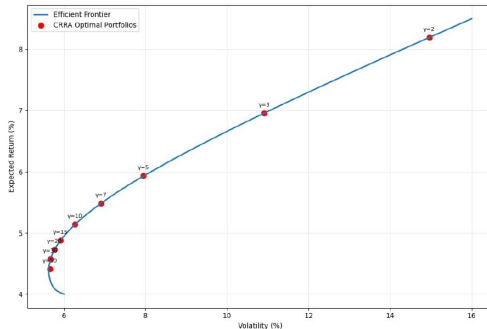
The MV approach is a special case of **expected utility maximisation**:

$$\max_{\mathbf{w}} \mathbb{E} [U(W_T)] \quad \text{s.t.} \quad \mathbf{1}'\mathbf{w} = 1$$

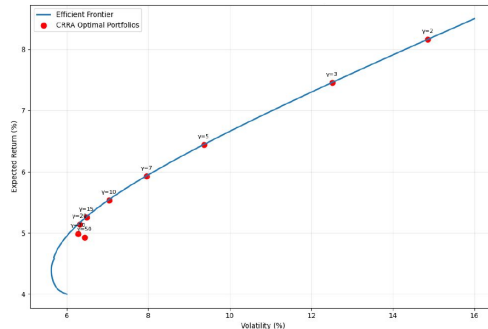
- Few studies provide evidence of material benefits of expected utility over MV (Adler and Kritzman, 2007; Sharpe, 2007)
- The most common alternative is **CRRA / power utility**
- In many cases MV provides a very good approximation to CRRA (Kroll, Levy and Markowitz, 1984)
- Greater risk aversion amplifies the difference between MV and CRRA

MV and CRRA efficient frontiers

normal distributions



non-normal distributions



Note: For more content, see **Jupyter Notebook exercise**

Benchmarks

The previous formulation allows a free choice of portfolio weights (subject to summing to one). However, this does not correspond to most portfolio managers' reality:

- Most managers have a **benchmark** portfolio that they are supposed to track
- An **active** manager can deviate from the benchmark, but usually only to a degree (an "active risk budget")
- Many managers face **long-only** constraints (each element of $\mathbf{w}$ must be weakly positive)
- Active managers may face explicit constraints on how far the chosen portfolio can deviate from the benchmark

We will discuss constraints in more depth later on.

Active positions and returns I

Most investment managers are given a benchmark portfolio against which their performance will be judged (e.g. a US large-cap manager may have the S&P 500 as a benchmark).

Defining $\mathbf{w}_b$ as a vector of benchmark weights, a manager's **active weights** ($\mathbf{w}_a$) are:

$$\mathbf{w}_a = \mathbf{w} - \mathbf{w}_b$$

The active weights:

- Represent the **deviations** from the benchmark that the manager has chosen in order to exploit her beliefs / information and so improve the risk-return trade-off in the portfolio
- **Sum to zero**, since they are the difference between two sets of weights that each sum to one

Active positions and returns II

Based on this decomposition, if R_t is the vector of security returns, we can write the active portfolio return in three equivalent ways:

$$R_{a,t} = R_{p,t} - R_{b,t} = \mathbf{w}'R_t - \mathbf{w}_b'R_t = \mathbf{w}_a'R_t$$

where $R_{p,t}$ is the total portfolio return and $R_{b,t}$ is the benchmark portfolio return.

Thus:

- The active return is the difference between the returns on two unit-investment portfolios.

Definition of benchmark beta

We first define the β of any portfolio relative to the benchmark portfolio:

$$\beta_p = \frac{\text{Cov}(R_{p,t} - R_{f,t}, R_{b,t} - R_{f,t})}{\text{Var}(R_{b,t} - R_{f,t})} \quad (1)$$

where $R_{f,t}$ is the risk-free rate.

Features of benchmark β :

- It is just a **linear regression** coefficient, similar in interpretation to CAPM beta but it does **not** rely on any of the CAPM assumptions or require you to believe in the CAPM.
- It measures the **sensitivity** of the portfolio's return to the benchmark return.
 - A positive (negative) beta means the portfolio moves with (against) the benchmark.
 - A beta of zero means the portfolio is insensitive to benchmark movements.

Definition of alpha

Analogous to the CAPM, we define the α for portfolio p relative to the benchmark:

$$\alpha_p = [\mathbb{E}(R_{p,t+1}) - R_{f,t+1}] - \beta_p \mathbb{E}(R_{b,t+1} - R_{f,t+1}) \quad (2)$$

where $R_{f,t+1}$ is the risk-free rate and $\mathbb{E}(R_{b,t+1} - R_{f,t+1})$ is the expected excess return on the benchmark.

Breakdown of expected portfolio return:

- **Time premium:** the reward for tying up funds for one period, equal to $R_{f,t+1}$.
- **Reward for bearing benchmark risk:** equal to $\beta_p \mathbb{E}(R_{b,t+1} - R_{f,t+1})$.
- **Portfolio α :** the extra return above the other two contributions. Positive α represents excess performance.

Goal: the PM wants to choose weights that deliver a portfolio with positive (average) α .

Ex-ante vs. ex-post alpha

So far, everything is based on **expectations** (i.e. forecasts), not realised outcomes.

- **Ex-ante α** : the **forecasted** difference between the portfolio's excess return and β times the expected market excess return.
- **Ex-post α** : the **realised** difference between the portfolio's actual excess return and β times the actual market excess return.

For compensation purposes:

- What matters is usually realised (ex-post) α .
- Managers choose portfolios with positive ex-ante α , hoping it translates into positive ex-post α .

Definition of residual risk

Residual risk is the variation in portfolio returns left over once benchmark co-movement has been stripped out.

Rewrite equation (2) as a regression equation by adding an error term ε_p and dropping expectation operators:

$$R_{p,t+1} - R_{f,t+1} = \alpha_p + \beta_p(R_{b,t+1} - R_{f,t+1}) + \varepsilon_{p,t+1} \quad (3)$$

We define residual risk (σ_p) as the **standard deviation of the error term**:

$$\sigma_p^2 = \text{Var}(\varepsilon_{p,t+1}) \quad (4)$$

Through the regression interpretation, σ_p^2 is the portion of return variation **not** explained by movements in the benchmark and instead arising from the manager's active positions.

Tracking error

- Managers are often evaluated on the basis of their portfolio's **tracking error**.
- Tracking error indicates how closely the portfolio follows its benchmark.

Defined as the **standard deviation of the difference between portfolio and benchmark returns**:

$$\begin{aligned} R_{p,t+1} - R_{b,t+1} &= \alpha_p + (\beta_p - 1)(R_{b,t+1} - R_{f,t+1}) + \varepsilon_{p,t+1} \\ TE_p^2 &= (\beta_p - 1)^2 \sigma_b^2 + \sigma_p^2 \end{aligned} \tag{5}$$

Tracking error therefore depends on:

- **Portfolio β** : β closer to one $\Rightarrow$ lower tracking error. What if $\beta = 1$?
- **Residual risk**: less residual risk $\Rightarrow$ lower tracking error.
- **Benchmark return variability**: beyond the manager's control.

Summary: benchmarking, alpha and residual risk

So, what we have at present:

- **Positive α :** excess performance of a portfolio over its benchmark.
- **Negative α :** under-performance of a portfolio relative to its benchmark.
- α can be defined relative to any benchmark one cares to choose.
- **Residual risk:** the part of portfolio return variation that is unrelated to the benchmark.

An active portfolio manager has a double objective:

1. Build a portfolio that has a big, positive α .
2. Simultaneously minimise residual risk.

Value added relative to benchmark: the objective function

Given a basket of possible assets, in order to choose an optimal portfolio we need an objective function. We define **value added** as:

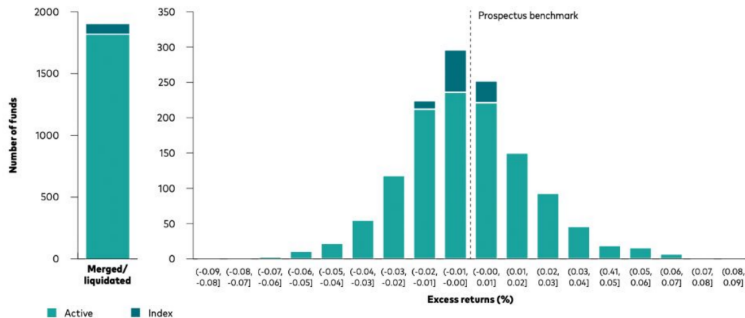
$$VA_p = \alpha_p - \lambda_r \sigma_p^2 \quad (6)$$

Comments:

- Analogous to mean-variance, but defined in terms of **active return** and **residual risk**: it measures active return adjusted for the active risk required to generate it.
- λ_r is the parameter capturing **aversion to residual risk**.
 - e.g. if $\lambda_r = 5$, residual risk of 5% requires an α of at least 1.25% for value added to be weakly positive.
- Curves of equal value added are parabolic in α - σ space: upward sloping and convex.

Do active equity managers add value?

Distribution of equity funds excess returns



Notes: Chart displays distribution of equity mutual funds' and ETFs' excess returns relative to their prospectus benchmarks for the 15 years to 31 December 2021. NAV-based performance; returns calculated in GBP, net of fees with income reinvested. Only funds available for sale in the UK are included.

Sources: "The case for low-cost index-fund investing", Vanguard, 2022.

Summary: every single input matters

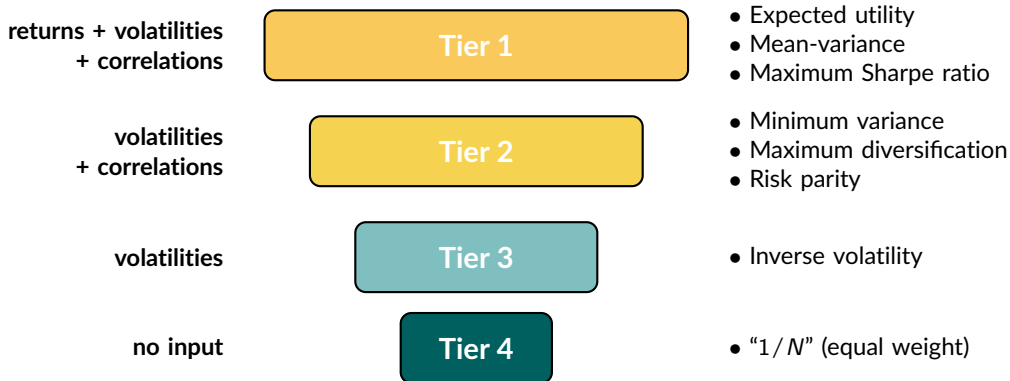
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(Treatment of costs is deferred – we return to it in Lecture 6.)

Many approaches to active management can add value

Different approaches require different **inputs** and **level of confidence in those inputs**.



The information ratio

Active managers often characterise their performance via a measure related to value added: the **information ratio**.

$$IR_p = \frac{\alpha_p}{\sigma_p} \quad (7)$$

Features:

- Like value added, the IR is increasing in α and decreasing in residual risk.
- IR depends on the measurement period of returns. By convention we report **annualised** information ratios.
- IRs are **independent of portfolio aggressiveness**: scaling all active weights up or down by a common factor leaves the IR unchanged.

Optimisation of value added for a given IR I

Assume that a particular manager is endowed with a particular IR .

- Each manager has her own IR – this is, implicitly, a statement about her forecasting skill.
- Any portfolio she builds inherits her IR .
- Thus the manager's IR defines our **investment opportunity set**.

Assuming an IR_p and maximising value added:

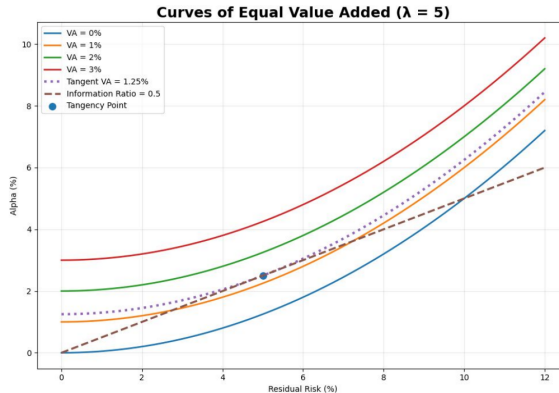
$$\max \alpha_p - \lambda_r \sigma_p^2 \quad \text{s.t.} \quad \frac{\alpha_p}{\sigma_p} = IR_p$$

We can rewrite this and solve for the optimal residual risk:

$$\max_{\sigma_p} IR_p \sigma_p - \lambda_r \sigma_p^2 \quad \Rightarrow \quad \sigma_p^* = \frac{IR_p}{2\lambda_r}$$

Optimisation of value added for a given IR II

Example: manager's $IR = 0.5$. The VA curves connect points of the same value added.



Optimisation of value added for a given IR III

At the optimal level of residual risk, value added simplifies to:

$$VA_p^* = \alpha_p - \lambda_r \sigma_p^2 = IR_p \sigma_p - \lambda_r \sigma_p^2 = \frac{IR_p^2}{2\lambda_r} - \lambda_r \frac{IR_p^2}{4\lambda_r^2} = \frac{IR_p^2}{4\lambda_r} \quad (8)$$

The amount of value the manager adds relative to the benchmark therefore depends on:

- The **information ratio**: more skilled managers add more value.
- Investor (active) **risk aversion**: the larger λ_r , the smaller the manager's scope for expressing skill.

From here we will treat risk aversion as fixed and ask, at the start of the next lecture, what determines a manager's IR.

Summary

In this lecture we have:

- Introduced a basic structure for thinking about **active management decisions**.
- Shown how to compute the **value added** for a portfolio and what determines it.
- Defined the **information ratio** and its relationship to portfolio α .

Next lecture: how to forecast returns – and so achieve a decent IR – and how to forecast and control risk effectively.